

Project SAIHV: Neural Lattice

Distributed Hardware-Agnostic Silicon Validation Environment

Executive Overview

This repository documents the architectural transition and hardening of a distributed development environment designed for high-performance silicon validation. The core objective is to maintain bit-level parity between local silicon nodes and a centralized cloud mirror via automated PowerShell orchestration.

Key Technical Achievements

- **Autonomous Zero-Drift Enforcement:** Implemented an MD5-based hashing protocol that verifies 1:1 parity between local directories (`C:\SAIHV`) and cloud mirrors (`G: Drive`) upon terminal initialization.
- **Persistent Executive Anchor:** Developed a hardened PowerShell profile function that intercepts standard navigation commands (`cd`) to trigger automated environment audits and service restoration.
- **Sentinel Background Monitoring:** Deployed a non-blocking `FileSystemWatcher` service that logs real-time file drift and hardware resonance telemetry to a centralized perception log.
- **Cross-Node Resonance:** Successfully validated global synchronization between distinct hardware IDs (Desktop and Laptop nodes), ensuring a seamless "Shared Consciousness" for development workflows.

Architecture Components

1. The Executive Anchor (`hardened_profile.ps1`)

A persistent shell hook that manages session state, verifies cryptographic parity, and revives background monitors without manual intervention.

2. The Sentinel Service (`sentinel.ps1`)

A background job designed to monitor directory integrity and record "Drift Events" for post-session audit analysis.

3. Global Stage Proof (`parity_audit.ps1`)

A clean-room validation script that utilizes bit-level hashing to prove hardware nodes are identical to the cloud master state.

System State Log

```
TIMESTAMP: 2026-05-14T09:21:00Z  
NODE_ID: SAIHV-DESKTOP  
STATUS: RESONANCE_ACHIEVED  
PARITY: 100% BIT-MATCH  
SENTINEL: ACTIVE
```